

```

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv("/content/archive.zip")

print(data.head())

data.rename(columns={'Gender Inequality Index (2021)': 'GII'},
            inplace=True)

mean = data['GII'].mean()
print("Mean:", mean)

median = data['GII'].median()
print("Median:", median)

var = data['GII'].var()
print("Variance:", var)

plt.hist(data['GII'], bins=20)
plt.xlabel("Gender Inequality Index")
plt.ylabel("Frequency")
plt.title("Distribution of Gender Inequality Index")
plt.show()

uniform_gii = np.random.uniform(
    low=data['GII'].min(),
    high=data['GII'].max(),
    size=len(data)
)

plt.hist(uniform_gii, bins=20)
plt.xlabel("Gender Inequality Index")
plt.ylabel("Frequency")
plt.title("Uniform Distribution of Gender Inequality Index")
plt.show()

print("Available columns:", data.columns.tolist())

```

|   | ISO3 | Country     | Continent | Hemisphere          | \ |
|---|------|-------------|-----------|---------------------|---|
| 0 | AFG  | Afghanistan | Asia      | Northern Hemisphere |   |
| 1 | AGO  | Angola      | Africa    | Southern Hemisphere |   |
| 2 | ALB  | Albania     | Europe    | Northern Hemisphere |   |
| 3 | AND  | Andorra     | Europe    | Northern Hemisphere |   |

|   |                                                                 |                      |                                  |                     |
|---|-----------------------------------------------------------------|----------------------|----------------------------------|---------------------|
| 4 | ARE                                                             | United Arab Emirates | Asia                             | Northern Hemisphere |
|   | Human Development Groups UNDP Developing Regions                |                      | HDI Rank (2021) \                |                     |
| 0 |                                                                 | Low                  | SA                               | 180.0               |
| 1 |                                                                 | Medium               | SSA                              | 148.0               |
| 2 |                                                                 | High                 | ECA                              | 67.0                |
| 3 |                                                                 | Very High            | NaN                              | 40.0                |
| 4 |                                                                 | Very High            | AS                               | 26.0                |
|   | GII Rank (2021) Gender Inequality Index (1990) \                |                      |                                  |                     |
| 0 |                                                                 | 167.0                | NaN                              |                     |
| 1 |                                                                 | 136.0                | 0.725                            |                     |
| 2 |                                                                 | 39.0                 | NaN                              |                     |
| 3 |                                                                 | NaN                  | NaN                              |                     |
| 4 |                                                                 | 11.0                 | 0.659                            |                     |
|   | Gender Inequality Index (1991) ...                              |                      | Gender Inequality Index (2012) \ |                     |
| 0 |                                                                 | NaN ...              |                                  | 0.738               |
| 1 |                                                                 | 0.723 ...            |                                  | 0.545               |
| 2 |                                                                 | NaN ...              |                                  | 0.235               |
| 3 |                                                                 | NaN ...              |                                  | NaN                 |
| 4 |                                                                 | 0.647 ...            |                                  | 0.171               |
|   | Gender Inequality Index (2013) Gender Inequality Index (2014) \ |                      |                                  |                     |
| 0 |                                                                 | 0.728                |                                  | 0.718               |
| 1 |                                                                 | 0.540                |                                  | 0.531               |
| 2 |                                                                 | 0.225                |                                  | 0.219               |
| 3 |                                                                 | NaN                  |                                  | NaN                 |
| 4 |                                                                 | 0.161                |                                  | 0.151               |
|   | Gender Inequality Index (2015) Gender Inequality Index (2016) \ |                      |                                  |                     |
| 0 |                                                                 | 0.706                |                                  | 0.692               |
| 1 |                                                                 | 0.530                |                                  | 0.529               |
| 2 |                                                                 | 0.204                |                                  | 0.191               |
| 3 |                                                                 | NaN                  |                                  | NaN                 |
| 4 |                                                                 | 0.126                |                                  | 0.118               |
|   | Gender Inequality Index (2017) Gender Inequality Index (2018) \ |                      |                                  |                     |
| 0 |                                                                 | 0.678                |                                  | 0.671               |
| 1 |                                                                 | 0.538                |                                  | 0.537               |
| 2 |                                                                 | 0.170                |                                  | 0.164               |
| 3 |                                                                 | NaN                  |                                  | NaN                 |
| 4 |                                                                 | 0.112                |                                  | 0.103               |

|   | Gender Inequality Index (2019) | Gender Inequality Index (2020) | \ |
|---|--------------------------------|--------------------------------|---|
| 0 | 0.665                          | 0.674                          |   |
| 1 | 0.537                          | 0.537                          |   |
| 2 | 0.156                          | 0.156                          |   |
| 3 | NaN                            | NaN                            |   |
| 4 | 0.056                          | 0.050                          |   |

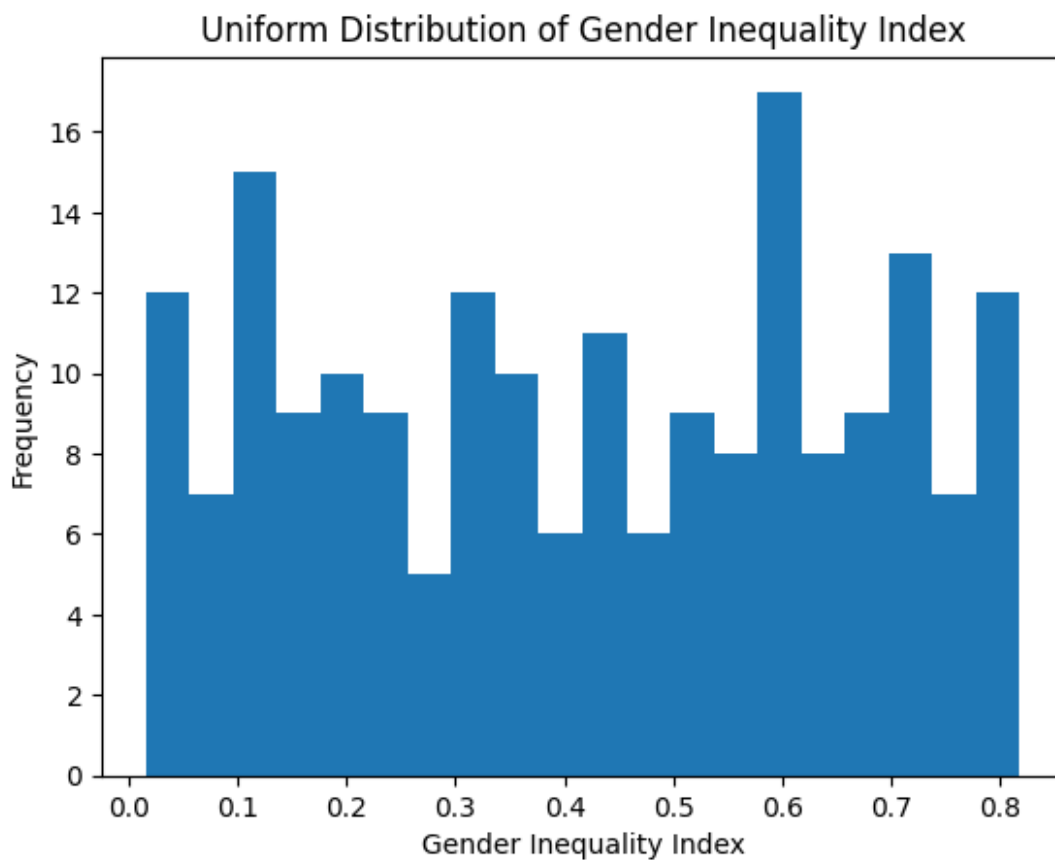
  

|   | Gender Inequality Index (2021) |
|---|--------------------------------|
| 0 | 0.678                          |
| 1 | 0.537                          |
| 2 | 0.144                          |
| 3 | NaN                            |
| 4 | 0.049                          |

[5 rows x 40 columns]  
Mean: 0.34437647058823534  
Median: 0.363  
Variance: 0.03885049648451099





Available columns: ['IS03', 'Country', 'Continent', 'Hemisphere', 'Human Development Groups', 'UNDP Developing Regions', 'HDI Rank (2021)', 'GII Rank (2021)', 'Gender Inequality Index (1990)', 'Gender Inequality Index (1991)', 'Gender Inequality Index (1992)', 'Gender Inequality Index (1993)', 'Gender Inequality Index (1994)', 'Gender Inequality Index (1995)', 'Gender Inequality Index (1996)', 'Gender Inequality Index (1997)', 'Gender Inequality Index (1998)', 'Gender Inequality Index (1999)', 'Gender Inequality Index (2000)', 'Gender Inequality Index (2001)', 'Gender Inequality Index (2002)', 'Gender Inequality Index (2003)', 'Gender Inequality Index (2004)', 'Gender Inequality Index (2005)', 'Gender Inequality Index (2006)', 'Gender Inequality Index (2007)', 'Gender Inequality Index (2008)', 'Gender Inequality Index (2009)', 'Gender Inequality Index (2010)', 'Gender Inequality Index (2011)', 'Gender Inequality Index (2012)', 'Gender Inequality Index (2013)', 'Gender Inequality Index (2014)', 'Gender Inequality Index (2015)', 'Gender Inequality Index (2016)', 'Gender Inequality Index (2017)', 'Gender Inequality Index (2018)', 'Gender Inequality Index (2019)', 'Gender Inequality Index (2020)', 'GII']